

# Lecture 4: Bifurcations and Qualitative Changes in Dynamics

Modelling Complex Systems (Spring 2026)

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## Code for Lecture 4

Have a look at the code `./scriptSStandardMap/ChirikovStandardMap.py` before Lecture 4.

Questions:

- What are the equations defining the map?
- What happens if we change the value of  $k$ ?
- What do you observe in the plots?

# Lecture goals

- Define what is meant by a parameter-dependent dynamical system.
- Explain what a bifurcation is.
- Relate bifurcations to stability changes of fixed points and periodic orbits.
- Study the main local bifurcations for one-dimensional discrete maps.
- Connect the theory with the logistic map and with numerical exploration of the standard map.

# Why parameters matter

# A parameter-dependent system

In applications, a dynamical system is rarely given by a single fixed rule.

We usually study a family of systems depending on a parameter  $\mu$ :

$$x_{n+1} = f_{\mu}(x_n).$$

The parameter may represent:

- a reproduction rate,
- a coupling strength,
- an external forcing,
- a discretization or control parameter.

# Qualitative versus quantitative change

Not every change in a parameter is equally important.

- A **quantitative change** modifies numerical values while preserving the same type of long-time behavior.
- A **qualitative change** modifies the structure of the dynamics itself.

Examples of qualitative change:

- a stable equilibrium becomes unstable,
- a new equilibrium appears,
- a stable periodic orbit is created,
- ordered motion gives way to more complicated behavior.

# Definition of bifurcation

## Informal definition

A **bifurcation** occurs when a small change in a parameter produces a qualitative change in the long-time dynamics.

What may change at a bifurcation value  $\mu = \mu_c$ ?

- the number of fixed points,
- the stability of a fixed point,
- the existence of periodic orbits,
- the geometry of invariant sets.

# What do we track as the parameter varies?

Typical objects of interest are:

- fixed points  $x^*(\mu)$ ,
- periodic orbits,
- their stability,
- the observed long-time attractors.

In practice, we often combine:

- analytic calculations,
- local linearization,
- numerical continuation,
- direct simulation.

# Local viewpoint

# Why does stability matter for bifurcations?

For a one-dimensional map

$$x_{n+1} = f_{\mu}(x_n),$$

a fixed point satisfies

$$f_{\mu}(x^*) = x^*.$$

Its local stability is determined by the derivative:

$$|f'_{\mu}(x^*)| < 1 \quad \text{attracting}, \quad |f'_{\mu}(x^*)| > 1 \quad \text{repelling}.$$

Therefore a bifurcation is often expected when

$$|f'_{\mu}(x^*)| = 1.$$

# Two codimension-one thresholds in 1D maps

In one-dimensional discrete dynamics, the most important local thresholds are:

- **Multiplier**  $+1$ : often associated with saddle-node or transcritical type bifurcations.
- **Multiplier**  $-1$ : often associated with period-doubling bifurcations.

Here the multiplier is simply

$$\lambda = f'_\mu(x^*).$$

# A warning

The condition

$$|f'_\mu(x^*)| = 1$$

is usually *necessary* for a local change of stability, but not by itself sufficient to classify the bifurcation.

To determine what really happens, we also need:

- how the fixed-point equation depends on the parameter,
- nonlinear terms beyond the linearization,
- sometimes numerical evidence.

# Main examples in one-dimensional maps

# Saddle-node bifurcation

Informally, two fixed points collide and disappear, or are created, as the parameter crosses a critical value.

Typical picture:

- for  $\mu < \mu_c$ : no fixed points,
- at  $\mu = \mu_c$ : one neutral fixed point,
- for  $\mu > \mu_c$ : two fixed points, usually one attracting and one repelling.

Canonical model:

$$x_{n+1} = x_n + \mu - x_n^2.$$

# Checking the saddle-node model

Fixed points satisfy

$$x = x + \mu - x^2 \quad \Longleftrightarrow \quad x^2 = \mu.$$

Thus:

- if  $\mu < 0$ , there are no real fixed points,
- if  $\mu = 0$ , there is one fixed point  $x^* = 0$ ,
- if  $\mu > 0$ , there are two fixed points  $x^* = \pm\sqrt{\mu}$ .

Since

$$f'_\mu(x) = 1 - 2x,$$

the two branches have different stability.

# Transcritical bifurcation

In a transcritical bifurcation, two fixed-point branches intersect and exchange stability.

Canonical model:

$$x_{n+1} = (1 + \mu)x_n - x_n^2.$$

Fixed points satisfy

$$(1 + \mu)x - x^2 = x \quad \Longleftrightarrow \quad x(\mu - x) = 0.$$

Therefore the branches are

$$x_1^* = 0, \quad x_2^* = \mu.$$

# The logistic map at $r = 1$

Recall the logistic map

$$x_{n+1} = rx_n(1 - x_n).$$

Its fixed points are

$$x_1^* = 0, \quad x_2^* = 1 - \frac{1}{r}.$$

At  $r = 1$ :

- the branch  $x^* = 0$  changes stability,
- the nonzero branch meets it,
- the long-time dynamics changes from extinction to attraction toward a positive equilibrium.

This is the basic transcritical-type mechanism in the logistic family.

# Period-doubling bifurcation

In a period-doubling bifurcation, a fixed point loses stability and a stable period-2 orbit appears.

The characteristic threshold is

$$f'_\mu(x^*) = -1.$$

Why period 2?

- one iterate sends a perturbation to the opposite side,
- the second iterate brings it back,
- so the natural object to inspect is  $f_\mu^2$ .

# The logistic map at $r = 3$

For the nonzero fixed point of the logistic map, we found

$$f'_r\left(1 - \frac{1}{r}\right) = 2 - r.$$

Therefore

$$2 - r = -1 \quad \Longleftrightarrow \quad r = 3.$$

Interpretation:

- for  $1 < r < 3$ , the nonzero fixed point is attracting,
- at  $r = 3$ , it becomes neutral,
- for  $r > 3$ , it is no longer attracting, and a stable period-2 orbit emerges.

# Qualitative changes seen in the bifurcation diagram

The bifurcation diagram of the logistic map records these changes visually.

Reading it from left to right:

- one stable fixed point,
- then a different stable fixed point after  $r = 1$ ,
- then a split into two values after  $r = 3$ ,
- then repeated period doublings,
- and eventually a broad region of chaotic behavior.

# Beyond one dimension

# Why higher-dimensional maps are richer

For a map in two or more dimensions,

$$z_{n+1} = F_\mu(z_n), \quad z_n \in \mathbb{R}^d,$$

local stability is determined by the eigenvalues of the Jacobian matrix.

This already allows phenomena that do not occur in one dimension:

- rotation combined with contraction or expansion,
- coexistence of regular and irregular regions,
- much more intricate invariant geometry.

# The Chirikov standard map

The prelecture code studies the standard map on the torus:

$$y_{n+1} = y_n + \frac{k}{2\pi} \sin(2\pi x_n) \pmod{1},$$

$$x_{n+1} = x_n + y_{n+1} \pmod{1}.$$

Here  $k$  controls the strength of the nonlinear perturbation.

Even without deriving a full bifurcation diagram, we can already ask how the phase portrait changes as  $k$  varies.

# What changes when $k$ changes?

Numerical exploration suggests the following qualitative picture:

- for small  $k$ , the motion is closer to regular organized patterns,
- as  $k$  increases, the geometry of the iterates becomes more distorted,
- different regions of phase space may show different behavior,
- the phase portrait reveals qualitative transitions that are harder to summarize with a single scalar diagram.

This is an important message for complex systems: not every parameter study is one-dimensional, and not every qualitative change is best represented by a simple branch diagram.

# Code used in this lecture

The Python script for today is

`scriptSStandardMap/ChirikovStandardMap.py.`

It does the following:

- choose many initial conditions on the torus
- for a fixed  $k$ , iterate the map forward
- plot all iterates in phase space
- change  $k$  with a slider and recompute

So the code is a tool for observing qualitative changes directly in the geometry of the orbits.

# Summary

# Take-away points

- A bifurcation is a qualitative change in long-time dynamics caused by a parameter variation.
- In one-dimensional maps, fixed-point bifurcations are often detected when the multiplier reaches  $+1$  or  $-1$ .
- The logistic map shows a transcritical-type change at  $r = 1$  and a period-doubling at  $r = 3$ .
- Bifurcation diagrams visualize how attracting long-time states depend on parameters.
- In higher-dimensional maps such as the standard map, qualitative transitions can be studied through phase portraits and numerical experiments.